

An hybrid method for phylogenetic network inference: application to the rice domestication



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Summary

Asian rice is one of the best documented crops in terms of genetic diversity. The domestication process, that probably started 9000 years ago in China, is hard to infer since this main vertical signal is inhibited by horizontal signal such as gene flow or introgressions. We propose to tackle here the reticulate evolution of rice sub-populations thanks to a rich stochastic model and to an hybrid method combining a classical phylogenetic network method and machine learning. We show, on simulated data, that our hybrid method enjoys very good performances. On rice real data, it infers a scenario with a unique domestication, the one of Japonica, followed by three reticulation events involving early Japonica. It highlights two introgression events at the origin of Indica and cAus, and one admixture event responsible of the emergence of cBas.

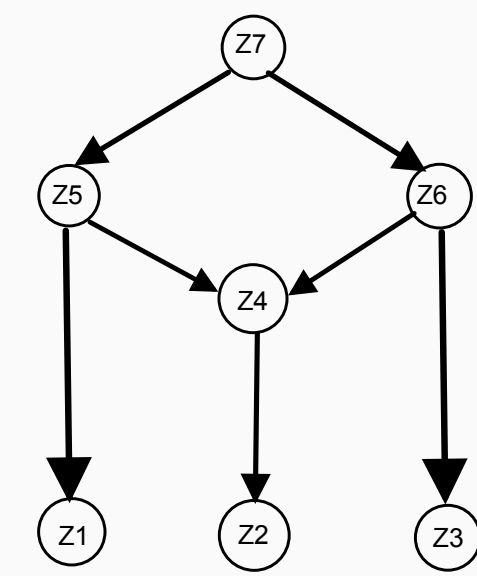
Phylogenetic networks

Phylogenetic networks are Directed Acyclic Graphs (DAG) that allow us to detect :

- hybridizations (e.g. plants)
- introgressions (e.g. plants and animals)
- horizontal gene transfer (e.g. bacteria)

Key points :

- Edge length = evolutionary time
- Dependencies between nodes
- Reticulation nodes have 2 parents and represent reticulation events



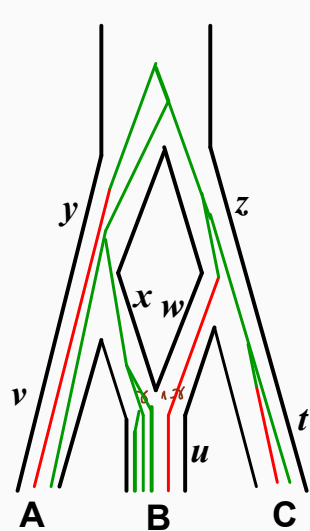
Our hybrid method

- a mathematical phylogenetic method : SNAPPNET (Rabier et al., PLoS Comput. Biol. 2021)
- a machine learning method : ABC-RF (Pudlo et al., Bioinform. 2015)

Mathematical model behind SNAPPNET

Network context

- Locus tree model (backward) :
 - Coalescent process
 - Yu et al. (Syst. Biol., 2011) model at the reticulation node
 - ⇒ Multispecies Network Coalescent
- SNP data model (forward)
 - Markov substitution model (two states : red or green)



- rA : number of red alleles in species A
- nA : number of individuals in species A
- rA, rB, rC are random
- nA, nB, nC are not random
- $Data = (rA, rB, rC)$
- Likelihood : $\mathbb{P}(Data | N)$ with N network

SNAPPNET : a Bayesian method for inferring network

- N : phylogenetic network (topology, branch lengths, population sizes, inheritance probability)
- X_i : data for locus i
- G_i : locus tree for locus i
- m loci

$$\mathbb{P}(N|Data) \propto \left(\prod_{i=1}^m \int_{\psi} \mathbb{P}(X_i|G_i) \mathbb{P}(G_i|S) dG_i \right) P(N) \\ \propto \mathbb{P}(Data | N) P(N)$$

SNAPPNET integrates over all locus trees (generalization of SNAPP, Bryant et al., Mol. Biol. Evol. 2012), using new algorithms dedicated to networks

Computation of the prior $P(N)$ by the birth hybridization process of Zhang et al. (Mol. Biol. Evol. 2018)

⇒ Markov Chain Monte Carlo (MCMC) in order to sample from the posterior distribution $\mathbb{P}(N|Data)$

Implemented within BEAST

ABC vs ABC-RF

ABC = Approximate Bayesian Computation (Rubin, Ann. Stat. 1984)

ABC-RF = ABC + Random Forest (Pudlo et al., Bioinform. 2015)

ABC-RF is more flexible than ABC

- Choice of Summary Statistics not as crucial as before
- ABC-RF does not rely on an accept-reject method
 - ⇒ Reference table can be filled easily (simulations from the prior)
- ABC-RF gives also the posterior probability of selecting the true model

Rice molecular diversity and our data-sets

899 accessions (Wang et al, Nature 2018) 29 selected accessions 64 selected accessions

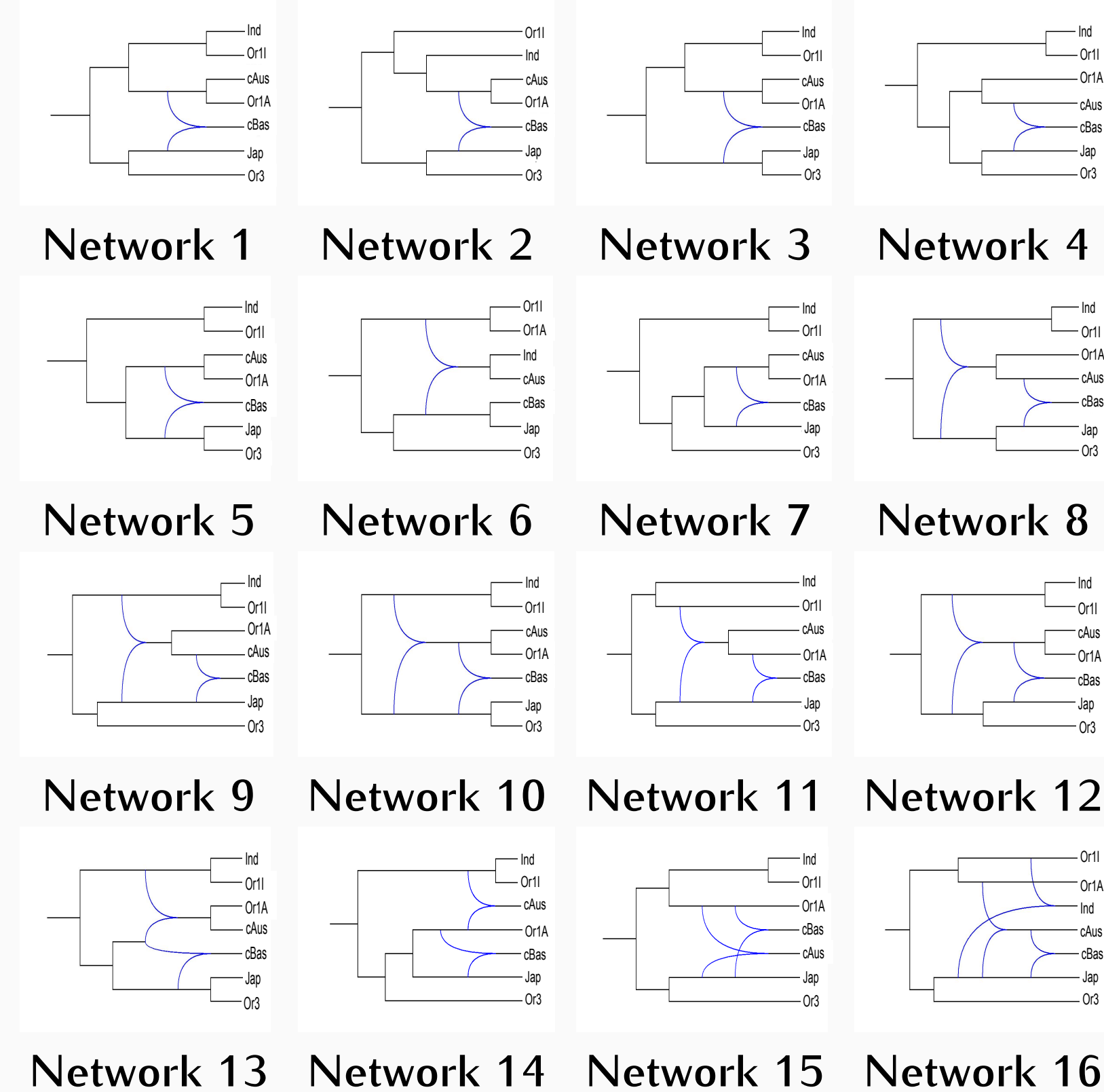


Unweighted Neighbour Joining (UWNJ) trees based on 2.48 million SNPs

Step 1 : SNAPPNET with penalized likelihood

4 Data-sets containing 18 accessions chosen from the 29 selected accessions
2 samplings of 12,000 markers along the whole genome

16 studied networks representing different evolutionary scenarios



6 months of computations

Network ID	Scanwidth	Average number of iterations
1	2	692,153
2	2	579,543
3	2	612,413
4	2	1,688,528
5	2	2,030,075
6	2	397,061
7	2	3,338,825
8	3	15,066
9	3	28,795
10	3	16,292
11	3	102,125
12	3	14,651
13	3	51,700
14	3	134,270
15	3	303,303
16	3	56,913

Networks 12, 8, 1, 15, 9, 16 are the preferred scenarios (AIC criterion)

SNAPPNET gives ML estimates of branch lengths, mutation rates, inheritance probabilities, scaled population sizes ...

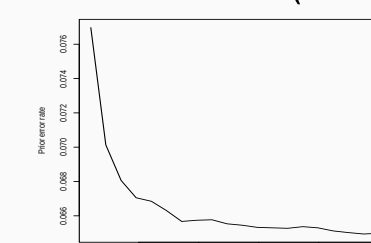
Step 2 : ABC-RF

Data-set with 64 accessions (2 samplings of 12,000 markers)

- 562 Summary Statistics
- 14,000 simulations per network
- Focus on the 6 best scenarios from step 1
- 8 combinations of priors based on SNAPPNET's estimates at step 1

Confusion matrix for prior 1						
	Net 1	Net 8	Net 9	Net 12	Net 15	Net 16
Net 1	10,565	290	57	3	94	28
Net 8	354	10,055	978	567	5	30
Net 9	75	1,399	10,330	197	25	27
Net 12	30	87	58	12,357	0	5
Net 15	45	6	9	1	11,709	93
Net 16	99	62	28	1	49	12,688
Error Rate	4.28%	16.13%	14.30%	1.44%	1.30%	1.85%

Error rate = f(nb trees)

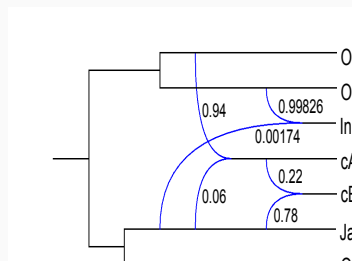


Results on real data

Prior ID	Sampling	Net 1	Net 8	Net 9	Net 12	Net 15	Net 16	Posterior Prob.
1	First	16.7%	2.6%	5.3%	6.2%	13.7%	55.5%	70.20%
	Second	18%	2.9%	5.8%	6.7%	14.8%	51.8%	67.64%
2	First	10.1%	1.6%	1.5%	1.4%	9%	76.4%	78.76%
	Second	12.1%	2.2%	1.7%	1.9%	10.3%	71.8%	80.90%
3	First	20.1%	3.7%	6.2%	4.5%	13.8%	51.7%	71.04%
	Second	23.0%	3.3%	6.7%	5.2%	14.9%	46.9%	70.17%
4	First	9.8%	1.6%	2.4%	1.1%	8.9%	76.2%	74.73%
	Second	10.8%	1.7%	2.8%	1.7%	11.3%	71.7%	73.31%
5	First	15.8%	0.2%	1.9%	8.2%	1.9%	72.0%	68.74%
	Second	16.0%	0.2%	1.4%	9.2%	1.9%	71.3%	68.05%
6	First	14.1%	0.1%	0.8%	2.3%	1.1%	81.6%	76.42%
	Second	14.3%	0.1%	0.6%	2.4%	1.9%	80.7%	76.52%
7	First	20.6%	0.1%	1.6%	10.4%	1.0%	66.3%	71.47%
	Second	20.9%	0.3%	1.3%	10.9%	1.1%	65.5%	69.22%
8	First	17.9%	0.1%	1.9%	4.7%	4.8%	70.6%	78.31%
	Second	18.9%	0.1%	2.1%	4.8%	4.6%	69.5%	79.02%

Conclusion : a unique domestication, Japonica

- Network 16 preferred scenario
- Two introgression events
- One admixture event



We retrieved a few domestication alleles from Japonica in Indica and cAus genomes